

A quadratic equation

The coefficients a , b , and c are real, with a nonzero.

$$ax^2 + bx + c = 0$$

Why must a be nonzero?

The zero-product property

A product equals zero exactly when at least one factor is zero.

$$uv = 0$$

Why does this property require the product to equal zero?

Solve a factored quadratic

$$(x - 5)(x + 2) = 0$$

What are the solutions?

Factor before solving

$$x^2 - 3x - 18 = 0$$

What are the solutions?

A nonunit leading coefficient

$$3x^2 - 5x - 2 = 0$$

How can we solve this quadratic by factoring?

Keep a zero solution

$$2x^2 = 10x$$

Which solutions could be lost by dividing by x ?

Make one side zero

$$x(x + 2) = 15$$

What are all solutions?

The square root property

For a positive real k , both square roots must be considered.

$$u^2 = k$$

Why can taking only the principal square root lose a solution?

Solve by square roots

$$(x + 4)^2 = 7$$

What are the exact real solutions?

A repeated root

$$(x - 3)^2 = 0$$

How many distinct solutions are there?

Complete a square

$$x^2 + 10x + c$$

What value of c makes this expression a perfect square?

Complete the square to solve

$$x^2 + 6x = 2$$

What exact solutions result from completing the square?

Normalize the leading coefficient

$$2x^2 - 8x + 1 = 0$$

How can completing the square solve this equation?

The quadratic formula

For real coefficients with a nonzero, solutions can be real or complex.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where do completing the square and this formula connect?

Identify the coefficients

$$5x^2 = 3x + 4$$

Which a, b, and c belong in the quadratic formula?

An irrational pair

$$2x^2 + x - 4 = 0$$

What are the exact solutions?

The discriminant

The discriminant is the expression under the radical in the quadratic formula.

$$D = b^2 - 4ac$$

How does its sign determine the kind of roots?

Predict the roots

$$4x^2 - 12x + 9 = 0$$

How many distinct real roots are there, before we solve?

A nonreal pair

$$x^2 + 2x + 5 = 0$$

What are its complex solutions?

Choose a method

$$7(x - 1)^2 = 28$$

Which solution method is especially efficient here?

Roots and a graph

$$y = x^2 - x - 6$$

How do its x-intercepts correspond to quadratic solutions?

Intersections with a line

$$y = x^2, \quad y = x + 6$$

What are the intersection points?

A rectangle model

A rectangle's length is 3 meters more than its width. Its area is 40 square meters.

What width is physically possible?

A vertical-motion model

Height h is in meters and time t in seconds.
Use $t \geq 0$.

$$h(t) = -5t^2 + 20t + 1$$

At what time does the object reach the ground?